

Garv Rawat

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SUMMARY

Computer Science student with experience in machine learning, natural language processing, data analysis, and full-stack web development. Proficient in building end-to-end applications involving data preprocessing, feature engineering, model training, evaluation, and deployment using Python, scikit-learn, and REST APIs.

Experienced in developing scalable web applications using React.js, Node.js, and MongoDB, and integrating machine learning models into production systems. Strong foundation in data structures, algorithms, and system design, with a focus on building efficient and data-driven solutions.

EDUCATION

Vellore Institute of Technology, Chennai

B.Tech in Electronics and Computer Engineering | CGPA: 9.06

Chennai, Tamil Nadu

2023 – Expected May 2027

Ryan International School, Greater Noida

Senior Secondary – Class 12 (CBSE) | Percentage: 92.6%

Greater Noida, Uttar Pradesh

2021 – 2022

Ryan International School, Greater Noida

Secondary – Class 10 (CBSE) | Percentage: 87.2%

Greater Noida, Uttar Pradesh

2019 – 2020

RELEVANT SKILLS

AI & Machine Learning: Scikit-learn, PyTorch, NLP, TF-IDF, Logistic Regression, Random Forest, Sentence Transformers, Model Evaluation (F1, Precision, Recall)

LLM & RAG: LangChain, ChromaDB, Groq (LLaMA 3), Retrieval-Augmented Generation, Semantic Search, Vector Embeddings

Web Development: React.js, Node.js, Express.js, Flask, REST APIs, Socket.io, HTML, CSS

Data Science: Pandas, NumPy, Matplotlib, EDA, Feature Engineering, Data Cleaning

Databases: MongoDB, MySQL, ChromaDB

Tools: Git, Docker, Joblib, VS Code, Vite, Jupyter Notebook

PROJECTS

DocChat – Multi-Document RAG Application | *Python, LangChain, ChromaDB, Flask, React.js, Groq*

- Built a multi-document RAG pipeline using LangChain and ChromaDB that enables semantic Q&A across multiple PDFs simultaneously, retrieving the top-3 most relevant chunks per query using cosine similarity over sentence-transformer embeddings (all-MiniLM-L6-v2).
- Implemented conversation memory with a 10-message sliding window, per-document chat history isolation, and source citations showing document name and page number for every answer.
- Deployed a 6-endpoint Flask REST API supporting PDF upload, document management, single-document chat, cross-document search, 2-document comparison, and history reset – integrated with a React.js frontend.

SocialSphere – AI-Powered Social Networking App | *React.js, Node.js, Python, Flask, Scikit-learn, Socket.io, MongoDB*

- Trained a TF-IDF + Logistic Regression sentiment classifier on 50,000 IMDB reviews achieving 88%+ accuracy (F1: 0.88), deployed via Flask REST API to automatically label social media posts as positive or negative in real time.
- Implemented real-time private messaging and instant follow notifications using Socket.io, supporting 50+ concurrent connections with sub-100ms event delivery.
- Built full-stack social platform with JWT authentication, image post feed, follow/unfollow system, and live online friends sidebar using React.js, Node.js, Express.js, and MongoDB.

CERTIFICATIONS & ACHIEVEMENTS

IBM Career Education | *IBM Developer Skills Network*

- DevOps Fundamentals – June 2025
- DevOps, Agile & Design Thinking – June 2025

Gujarat Robofest 4.0 | *National Level Competition*

- Secured **3rd place nationally** in the Underwater and Submarine category, awarded **INR 5,00,000**.
- Recognised for innovation in autonomous underwater robotics.

CO-CURRICULAR ACTIVITIES

- Deputy Lead of CAD and Simulations Department of a special robotics team.
- Actively trade forex and cryptocurrency markets using technical and fundamental analysis.